

Package ‘TruncatedPCQM’

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Title Density Estimation for Point-Centered Quarter Method with Truncated Sampling

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Description Implements a systematic methodology for estimating population density from point-centered quarter method (PCQM) surveys when distance measurements are truncated by a maximum search radius (right-censored). The package provides a unified framework for analyzing such incomplete data, addressing both completely randomly distributed (Poisson) and spatially aggregated (Negative Binomial) populations. Key features include:

- (1) Adjusted moment-based density estimators for censored distances;
- (2) Maximum likelihood estimation (MLE) of density under the Poisson (CSR) model; and
- (3) Simultaneous MLE of density and an aggregation parameter under the Negative Binomial model.

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Language en

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Depends R (>= 3.5.0)

Suggests testthat (>= 3.0.0), dplyr (>= 1.1.0)

Config/testthat/edition 3

URL <https://github.com/rcjhrpyt-droid/TruncatedPCQM>

BugReports <https://github.com/rcjhrpyt-droid/TruncatedPCQM/issues>

NeedsCompilation no

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adjusted_moments

*Adjusted Moment-based Estimators for Right-censored PCQM Data***Description**

Computes various density estimators (DK, Cottam, Pollard, Shen, Morisita) using adjusted moments to account for right-censoring in PCQM data.

Usage

```
adjusted_moments(
  distances,
  C = 20,
  q = 4,
  l = 1,
  init_method = "Pollard_censored"
)
```

Arguments

distances	A numeric vector, matrix, data.frame, or list of point-to-tree distances. For matrix/data.frame inputs, each row represents a sampling point and each column a sector.
C	Maximum search radius (censoring threshold). Distances exceeding C are treated as right-censored observations. Default is 20.
q	Number of sectors per sampling point. Default is 4.
l	Neighbor order. Default is 1.
init_method	Method for obtaining initial λ estimate used in NBD-based estimators. One of: "Pollard_censored" (default), "Cottam_censored", or "DK_censored" (only for $l = 1$).

Value

An object of class "pcqm_moments" containing:

- DK_censored: Diggle-Koedam estimator value (NA if l is not 1)
- Cottam_censored: Cottam-type estimator value
- Pollard_censored: Pollard-type estimator value
- Shen_censored: Shen's NBD-based estimator value (NA if estimated $k < 1$)
- Morisita_censored: Morisita's first estimator value (NA if $l \neq 1$)
- k_hat: Estimated aggregation parameter k from Shen's estimator
- censored_rate: Proportion of censored sectors
- n_points: Number of sampling points
- n_sectors: Total number of sectors analyzed
- n_censored: Number of censored observations ($R > C$)
- call: Original function call

Examples

```

distances_matrix <- matrix(c(
  9999.000000, 9999.000000, 9999.000000, 9999.000000,
  9999.000000, 7.655136, 13.815876, 10.423496,
  6.094721, 4.135461, 7.732912, 5.454545,
  9999.000000, 9999.000000, 14.787289, 15.670821,
  9999.000000, 9.825537, 11.611850, 15.757861,
  9999.000000, 9.670381, 14.055394, 17.075678,
  11.529219, 4.464136, 7.793114, 11.309553,
  13.307828, 5.864490, 13.309636, 5.897720,
  9999.000000, 9999.000000, 9999.000000, 9999.000000,
  9999.000000, 9999.000000, 18.201084, 9999.000000,
  9999.000000, 7.809056, 12.612496, 5.601366,
  9.201294, 9999.000000, 8.353524, 9.683701,
  6.592604, 19.117869, 19.758384, 12.923507,
  15.574824, 10.643719, 9.494539, 7.382031,
  9.143077, 9999.000000, 15.551414, 5.266916,
  9999.000000, 9999.000000, 9999.000000, 9999.000000,
  18.604278, 7.279454, 9.385355, 5.573127,
  9999.000000, 9999.000000, 9999.000000, 9999.000000,
  9999.000000, 9999.000000, 9999.000000, 9999.000000,
  11.980811, 9999.000000, 11.853695, 14.405252
), nrow = 20, ncol = 4, byrow = TRUE)

moment_result <- adjusted_moments(
  distances = distances_matrix,
  C = 20,
  q = 4,
  l = 2,
  init_method = "Pollard_censored"
)

moment_result

```

csr_mle

CSR-based Maximum Likelihood Estimator for Right-censored PCQM Data

Description

Computes the maximum likelihood estimate (MLE) of population density under complete spatial randomness (CSR), accounting for right-censored point-centered quarter method (PCQM) distances.

Usage

```

csr_mle(
  distances,
  C = 20,
  q = 4,

```

```

l = 1,
lambda_lower = 1e-04,
lambda_upper = 1
)

```

Arguments

distances	A numeric vector, matrix, data.frame, or list of point-to-tree distances. For matrix/data.frame inputs, each row represents a sampling point and each column a sector.
C	Maximum search radius (censoring threshold). Distances exceeding C are treated as right-censored observations. Default is 20.
q	Number of sectors per sampling point. Default is 4.
l	Neighbor order. Default is 1.
lambda_lower	Lower bound for λ in numerical optimization. Must be positive. Default is 1e-4.
lambda_upper	Upper bound for λ in numerical optimization. Must be positive. Default is 1.

Value

An object of class "pcqm_csr_mle" containing:

- lambda: MLE of population density (individuals per unit area)
- logLik: Maximized log-likelihood value
- n_obs: Number of fully observed (non-censored) distances
- n_censored: Number of censored observations ($R > C$)
- n_sectors: Total number of sectors analyzed
- n_points: Inferred number of sampling points
- censored_rate: Proportion of censored sectors
- call: Original function call

Examples

```

distances_matrix <- matrix(c(
  9999.000000, 9999.000000, 9999.000000, 9999.000000,
  9999.000000, 7.655136, 13.815876, 10.423496,
  6.094721, 4.135461, 7.732912, 5.454545,
  9999.000000, 9999.000000, 14.787289, 15.670821,
  9999.000000, 9.825537, 11.611850, 15.757861,
  9999.000000, 9.670381, 14.055394, 17.075678,
  11.529219, 4.464136, 7.793114, 11.309553,
  13.307828, 5.864490, 13.309636, 5.897720,
  9999.000000, 9999.000000, 9999.000000, 9999.000000,
  9999.000000, 9999.000000, 18.201084, 9999.000000,
  9999.000000, 7.809056, 12.612496, 5.601366,
  9.201294, 9999.000000, 8.353524, 9.683701,
  6.592604, 19.117869, 19.758384, 12.923507,
  15.574824, 10.643719, 9.494539, 7.382031,
  9.143077, 9999.000000, 15.551414, 5.266916,
  9999.000000, 9999.000000, 9999.000000, 9999.000000,
  18.604278, 7.279454, 9.385355, 5.573127,
  9999.000000, 9999.000000, 9999.000000, 9999.000000,

```

```

9999.000000, 9999.000000, 9999.000000, 9999.000000,
11.980811, 9999.000000, 11.853695, 14.405252
), nrow = 20, ncol = 4, byrow = TRUE)

csr_result <- csr_mle(
  distances = distances_matrix,
  C = 20,
  q = 4,
  l = 2,
  lambda_lower = 1e-4,
  lambda_upper = 1
)

csr_result

```

nbd_mle	<i>NBD-based Maximum Likelihood Estimator for Right-censored PCQM Data</i>
---------	--

Description

Computes the maximum likelihood estimate (MLE) of population density under Negative Binomial model, accounting for right-censored point-centered quarter method (PCQM) distances.

Usage

```

nbd_mle(
  distances,
  C = 20,
  q = 4,
  l = 1,
  init_method = "Shen_censored",
  lambda_lower = 1e-04,
  lambda_upper = 1
)

```

Arguments

distances	A numeric vector, matrix, data.frame, or list of point-to-tree distances. For matrix/data.frame inputs, each row represents a sampling point and each column a sector.
C	Maximum search radius (censoring threshold). Distances exceeding C are treated as right-censored observations. Default is 20.
q	Number of sectors per sampling point. Default is 4.
l	Neighbor order. Default is 1.
init_method	Method for obtaining initial parameter values. One of: "Shen_censored" (default), "Pollard_censored", "Cottam_censored", or "DK_censored" (l=1 only).
lambda_lower	Lower bound for λ in numerical optimization. Must be positive. Default is 1e-4.
lambda_upper	Upper bound for λ in numerical optimization. Must be positive. Default is 1.

Value

An object of class "pcqm_nbd_mle" containing:

- `lambda`: MLE of population density (individuals per unit area)
- `k`: MLE of aggregation parameter (smaller values indicate stronger aggregation)
- `logLik`: Maximized log-likelihood value
- `n_obs`: Number of fully observed (non-censored) distances
- `n_censored`: Number of censored observations ($R > C$)
- `n_sectors`: Total number of sectors analyzed
- `n_points`: Inferred number of sampling points
- `censored_rate`: Proportion of censored sectors
- `call`: Original function call

Examples

```
distances_matrix <- matrix(c(
  9999.000000, 9999.000000, 9999.000000, 9999.000000,
  9999.000000, 7.655136, 13.815876, 10.423496,
  6.094721, 4.135461, 7.732912, 5.454545,
  9999.000000, 9999.000000, 14.787289, 15.670821,
  9999.000000, 9.825537, 11.611850, 15.757861,
  9999.000000, 9.670381, 14.055394, 17.075678,
  11.529219, 4.464136, 7.793114, 11.309553,
  13.307828, 5.864490, 13.309636, 5.897720,
  9999.000000, 9999.000000, 9999.000000, 9999.000000,
  9999.000000, 9999.000000, 18.201084, 9999.000000,
  9999.000000, 7.809056, 12.612496, 5.601366,
  9.201294, 9999.000000, 8.353524, 9.683701,
  6.592604, 19.117869, 19.758384, 12.923507,
  15.574824, 10.643719, 9.494539, 7.382031,
  9.143077, 9999.000000, 15.551414, 5.266916,
  9999.000000, 9999.000000, 9999.000000, 9999.000000,
  18.604278, 7.279454, 9.385355, 5.573127,
  9999.000000, 9999.000000, 9999.000000, 9999.000000,
  9999.000000, 9999.000000, 9999.000000, 9999.000000,
  11.980811, 9999.000000, 11.853695, 14.405252
), nrow = 20, ncol = 4, byrow = TRUE)

nbd_result <- nbd_mle(
  distances = distances_matrix,
  C = 20,
  q = 4,
  l = 2,
  init_method = "Pollard_censored",
  lambda_lower = 1e-4,
  lambda_upper = 10
)

nbd_result
```

print

*Print Methods for PCQM Density Estimation Objects***Description**

Provides formatted console output for fitted PCQM (Point-Centered Quarter Method) density estimation results under different model assumptions. The following specialized print methods for S3 classes are supported:

- `print.pcqmqcsr_mle` – display CSR-based PCQM maximum likelihood estimation results.
- `print.pcqmqmoments` – display adjusted moment-based PCQM density estimators.
- `print.pcqmqnbd_mle` – display NBD-based PCQM maximum likelihood estimation results.

Usage

```
## S3 method for class 'pcqm_csr_mle'
print(x, digits = 6, ...)

## S3 method for class 'pcqm_moments'
print(x, digits = 6, ...)

## S3 method for class 'pcqm_nbd_mle'
print(x, digits = 6, ...)
```

Arguments

<code>x</code>	An object of class "pcqm_csr_mle", "pcqm_moments", or "pcqm_nbd_mle", returned by csr_mle , adjusted_moments , or nbd_mle , respectively.
<code>digits</code>	Number of decimal places to display for numeric estimates. Default is 6.
<code>...</code>	Additional arguments passed to the generic print method (currently unused; included for S3 consistency).

See Also

[csr_mle](#) for CSR-based MLE estimation; [adjusted_moments](#) for adjusted moment estimators; [nbd_mle](#) for NBD-based MLE estimation.

Examples

```
distances_matrix <- matrix(c(
  9999.000000, 9999.000000, 9999.000000, 9999.000000,
  9999.000000, 7.655136, 13.815876, 10.423496,
  6.094721, 4.135461, 7.732912, 5.454545,
  9999.000000, 9999.000000, 14.787289, 15.670821,
  9999.000000, 9.825537, 11.611850, 15.757861,
  9999.000000, 9.670381, 14.055394, 17.075678,
  11.529219, 4.464136, 7.793114, 11.309553,
  13.307828, 5.864490, 13.309636, 5.897720,
  9999.000000, 9999.000000, 9999.000000, 9999.000000,
```

```

9999.000000, 9999.000000, 18.201084, 9999.000000,
9999.000000, 7.809056, 12.612496, 5.601366,
9.201294, 9999.000000, 8.353524, 9.683701,
6.592604, 19.117869, 19.758384, 12.923507,
15.574824, 10.643719, 9.494539, 7.382031,
9.143077, 9999.000000, 15.551414, 5.266916,
9999.000000, 9999.000000, 9999.000000, 9999.000000,
18.604278, 7.279454, 9.385355, 5.573127,
9999.000000, 9999.000000, 9999.000000, 9999.000000,
9999.000000, 9999.000000, 9999.000000, 9999.000000,
11.980811, 9999.000000, 11.853695, 14.405252
), nrow = 20, ncol = 4, byrow = TRUE)

# CSR-based MLE
csr_result <- csr_mle(
  distances = distances_matrix,
  C = 20,
  q = 4,
  l = 2,
  lambda_lower = 1e-4,
  lambda_upper = 1
)
print(csr_result)

# Adjusted Moment Estimators
moment_result <- adjusted_moments(
  distances = distances_matrix,
  C = 20,
  q = 4,
  l = 2,
  init_method = "Pollard_censored"
)
print(moment_result)

# NBD-based MLE
nbd_result <- nbd_mle(
  distances = distances_matrix,
  C = 20,
  q = 4,
  l = 2,
  init_method = "Pollard_censored",
  lambda_lower = 1e-4,
  lambda_upper = 10
)
print(nbd_result)

```

Description

Generates structured summary objects and formatted console output for PCQM (Point-Centered Quarter Method) density estimation results under different model assumptions, with detailed statistical information beyond basic print output. The following specialized S3 methods are supported:

- `summary.pcq_mcsr_mle` – create summary object for CSR-based PCQM MLE results
- `summary.pcq_mmoments` – create summary object for adjusted moment-based PCQM estimators
- `summary.pcq_mnbd_mle` – create summary object for NBD-based PCQM MLE results
- `print.summary.pcq_mcsr_mle` – formatted print for CSR-based PCQM MLE summary
- `print.summary.pcq_mmoments` – formatted print for adjusted moment estimator summary
- `print.summary.pcq_mnbd_mle` – formatted print for NBD-based PCQM MLE summary

Usage

```
## S3 method for class 'pcq_mcsr_mle'
summary(object, ...)

## S3 method for class 'summary.pcq_mcsr_mle'
print(x, digits = 6, ...)

## S3 method for class 'pcq_mmoments'
summary(object, ...)

## S3 method for class 'summary.pcq_mmoments'
print(x, digits = 6, ...)

## S3 method for class 'pcq_mnbd_mle'
summary(object, ...)

## S3 method for class 'summary.pcq_mnbd_mle'
print(x, digits = 6, ...)
```

Arguments

<code>object</code>	An object of class "pcq_mcsr_mle", "pcq_mmoments", or "pcq_mnbd_mle", returned by <code>csr_mle</code> , <code>adjusted_moments</code> , or <code>nbd_mle</code> , respectively. Used for the <code>summary.*</code> S3 methods only.
<code>...</code>	Additional arguments passed to generic <code>summary</code> or <code>print</code> methods (currently unused; included for S3 consistency).
<code>x</code>	An object of class "summary.pcq_mcsr_mle", "summary.pcq_mmoments", or "summary.pcq_mnbd_mle", returned by the corresponding <code>summary.*</code> method. Used for the <code>print.summary.*</code> S3 methods only.
<code>digits</code>	Number of decimal places to display for numeric estimates. Default is 6.

See Also

`csr_mle` for CSR-based MLE estimation; `adjusted_moments` for adjusted moment estimators; `nbd_mle` for NBD-based MLE estimation; `print` for basic PCQM estimation result display.

Examples

```
distances_matrix <- matrix(c(
  9999.000000, 9999.000000, 9999.000000, 9999.000000,
  9999.000000, 7.655136, 13.815876, 10.423496,
  6.094721, 4.135461, 7.732912, 5.454545,
```

```

9999.000000, 9999.000000, 14.787289, 15.670821,
9999.000000, 9.825537, 11.611850, 15.757861,
9999.000000, 9.670381, 14.055394, 17.075678,
11.529219, 4.464136, 7.793114, 11.309553,
13.307828, 5.864490, 13.309636, 5.897720,
9999.000000, 9999.000000, 9999.000000, 9999.000000,
9999.000000, 9999.000000, 18.201084, 9999.000000,
9999.000000, 7.809056, 12.612496, 5.601366,
9.201294, 9999.000000, 8.353524, 9.683701,
6.592604, 19.117869, 19.758384, 12.923507,
15.574824, 10.643719, 9.494539, 7.382031,
9.143077, 9999.000000, 15.551414, 5.266916,
9999.000000, 9999.000000, 9999.000000, 9999.000000,
18.604278, 7.279454, 9.385355, 5.573127,
9999.000000, 9999.000000, 9999.000000, 9999.000000,
9999.000000, 9999.000000, 9999.000000, 9999.000000,
11.980811, 9999.000000, 11.853695, 14.405252
), nrow = 20, ncol = 4, byrow = TRUE)

# CSR-based MLE Summary
csr_result <- csr_mle(distances = distances_matrix, C = 20, q = 4, l = 2,
                      lambda_lower = 1e-4, lambda_upper = 1)
summary(csr_result)

# Adjusted Moment Estimators Summary
moment_result <- adjusted_moments(distances = distances_matrix, C = 20, q = 4, l = 2,
                                   init_method = "Pollard_censored")
summary(moment_result)

# NBD-based MLE Summary
nbd_result <- nbd_mle(distances = distances_matrix, C = 20, q = 4, l = 2,
                      init_method = "Pollard_censored", lambda_lower = 1e-4, lambda_upper = 10)
summary(nbd_result)

```

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